

Task 3 - System Programming / File I/O Report

Objective

To demonstrate file handling in C++ by creating a file, writing data into it, reading the data back, and handling file errors gracefully.

Project Description

This project demonstrates basic File I/O operations using C++. The program creates a text file named student.txt, writes student details into the file, closes it, then reopens the file, reads the contents, and displays them on the console.

Files Included

- file_io.cpp
- student.txt
- README.md

Compilation Command

```
g++ file_io.cpp -o file_io
```

Run Command

```
.\file_io.exe
```

Expected Output

Data written successfully.

Contents of the file:

Name: Kalpana

Department: ECE

College: RMD Engineering College

Time Complexity

Writing to File : $O(n)$

Reading from File : $O(n)$

Screenshots

```
file_io.cpp > main()
1  #include <iostream>
2  #include <fstream>
3  #include <string>
4
5  using namespace std;
6
7  int main()
8  {
9      ofstream outFile("student.txt");
10
11     if (!outFile)
12     {
13         cout << "Error: Unable to create file!" << endl;
14         return 1;
15     }
16
17     outFile << "Name: Kalpana" << endl;
18     outFile << "Department: ECE" << endl;
19     outFile << "College: RMD Engineering College" << endl;
20
21     outFile.close();
22
23     cout << "Data written successfully." << endl;
24
25     ifstream inFile("student.txt");
26
27     if (!inFile)
28     {
29         cout << "Error: Unable to open file!" << endl;
30         return 1;
31     }
32
33     string line;
34
35     cout << "\nContents of the file:\n";
36
37     while (getline(inFile, line))
38     {
39         cout << line << endl;
40     }
41
42     inFile.close();
43
44     return 0;
45 }
```

```
student.txt
1  Name: Kalpana
2  Department: ECE
3  College: RMD Engineering College
4
```

Conclusion

The project successfully demonstrates file creation, writing, reading, and error handling using C++. The output verifies that File I/O operations work correctly.